RadioML Modulation Classification: Progress Report

Status: in progress. Data checks and filtering are done, feature extraction has started, and the classifier is not built. There is no accuracy result yet. Values read from plots are approximate.

1 Problem and Approach

Automatic modulation classification (AMC) identifies the modulation scheme of a received signal before demodulation. This project uses a classical pipeline (filtering, hand-designed features, simple classifier) so that every step can be explained and justified.

The data is RadioML 2016.10a: 11 modulations (8PSK, AM-DSB, AM-SSB, BPSK, CPFSK, GFSK, PAM4, QAM16, QAM64, QPSK, WBFM), 20 SNR levels from -20 to $+18$ dB, 1000 examples per modulation and SNR, each with 128 complex I/Q samples. Tools: Python (NumPy, SciPy, Matplotlib).

2 Work Completed

Constellations. Raw I/Q plots at 18 dB show a compact blob for AM-DSB and winding trails for QPSK and QAM16, not clean clusters. This is consistent with multi-sample symbols and uncorrected carrier phase drift; fixing it needs synchronization, which is out of scope.

Filter design. Two low-pass filters were compared: a 51-tap windowed FIR and a 4th-order Butterworth IIR. The FIR has constant group delay (25 samples) but a ripple floor near -55 to -60 dB. The IIR rolls off more gradually and its group delay varies from about 3 to 5.4 samples, so it distorts phase slightly.

Cutoff from measured spectra. Welch spectra (with `detrend=False`, after removing a spurious dip at 0 Hz caused by SciPy's default mean removal) show that the digital modulations (8PSK, BPSK, PAM4, QAM16, QAM64, QPSK) occupy about ± 0.1 to ± 0.12 cycles/sample. The initial guess of 0.25 of Nyquist sat at the edge of this band, so the shared cutoff was set to **0.3 of Nyquist** (± 0.15 cycles/sample). One filter is used for all classes because the modulation is unknown at filtering time.

Effect of filtering. On 1000 QPSK examples at 0 dB (first 64 samples dropped to avoid the FIR start-up transient), out-of-band noise fell from about -47 to -110 dB (FIR), with in-band content unchanged. Total power removed is about 5.0–5.3 dB at very low SNR, matching the 5.2 dB expected for white noise with 30% of the band kept, and only 0.1–0.2 dB at high SNR. Both filters differ by only 0.2–0.3 dB.

Reliability of single examples. QPSK example 0 was nearly silent for about 75 samples. Checks at 10 dB show average power is flat across the window, but a few examples are strongly imbalanced between window halves (more than 3 dB: 7/1000 QPSK, 6/1000 BPSK, 28/1000 QAM16, none for AM-DSB, AM-SSB, WBFM). Example 0 was the worst of 1000 (23.2 dB). Classes are therefore judged from all 1000 examples, never one plot.

3 Findings So Far

- Raw constellations separate amplitude-only modulation (blob) from phase-varying modulation (trails) even without synchronization.
- Filtering removes only out-of-band noise; for white noise the SNR gain is capped near 5 dB here.
- The FIR preserves timing (linear phase) at higher cost; the IIR is cheaper but distorts phase slightly.
- Fewer than 3% of QPSK, BPSK and QAM16 examples have most power in one half of the window.

4 Next Steps

Extract features over all examples, build the classifier, and evaluate accuracy versus SNR.